Classes and objects

Module 3 - Lesson 1

Overview

Object-oriented programming groups data and the functions that operate on it into classes. Instances of a class carry their own state and behavior.

Key Concepts

- class defines a blueprint; __init__ sets up each new instance.
- Methods are functions defined on the class and receive self as the instance.
- Instance attributes live on self; class attributes are shared by all instances.
- Property descriptors let you compute attributes and add validation.
- __str__ and __repr__ control how objects print and display.
- Encapsulation hides internal details behind a public interface.

Example

```
class BankAccount:
    def __init__(self, owner, balance=0):
        self.owner = owner
        self._balance = balance

    def deposit(self, amount):
        self._balance += amount
        return self._balance

    def __str__(self):
        return f"{self.owner}: ${self._balance}"
```

Summary

Classes bundle state and behavior into reusable objects. __init__ sets up instances, methods operate on them, and dunder methods integrate with Python’s syntax.

Practice Checklist

- Define a class with __init__, two methods, and a __str__.
- Create several instances and verify they keep independent state.
- Add a property with validation to one of the attributes.